

FZ-VIS Hands On

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Install FZ-VIS Locally (Method 1)

- Download FZ-VIS from GitHub

```
$ git clone https://github.com/YuxiaoLi1234/fzvis.git
```

- Run the installation script

```
$ ./scripts/install_fzvis.sh
```

```
============================================================
 1. Spack Setup
============================================================
[INFO] Existing Spack installation detected at: /home/guoxil/s
Use existing Spack and skip downloading/updating Spack? [Y/n]

Checking Libpressio Spack package repository...
[INFO] Libpressio repository already detected at: /home/guoxi
Skip updating Libpressio Spack package repository? [Y/n]

============================================================
 2. Spack Environment Configuration (fzvis-env)
============================================================
Please choose an option for the Spack environment:
  1) Create a new Spack environment
  2) Use an existing Spack environment
[INFO] Detected existing environment candidate: /home/guoxil/s
Select option [1/2] [default: 2]: 2
Enter the location of the existing Spack environment [default
[INFO] Verified existing Spack environment at: /home/guoxil/so
Activating Spack environment at /home/guoxil/software/spack/va
[INFO] Configuration memorized in /home/guoxil/software/spack/

============================================================
 3. Spack Dependencies (libpressio, python, node-js)
============================================================
[INFO] libpressio is ALREADY INSTALLED in this environment: li
Skip adding and installing Spack dependencies (libpressio, et
[INFO] Skipping Spack package installation.
```

```
============================================================
 4. Python Requirements (requirements.txt)
============================================================
[INFO] All required Python packages from requirements.txt are
Skip running 'pip install -r requirements.txt'? [Y/n]
[INFO] Skipping Python pip installation.

============================================================
 5. Frontend Dependencies & Build
============================================================
[INFO] Production frontend build ALREADY EXISTS at /home/guoxi
Skip frontend dependency installation and build? [Y/n]
[INFO] Skipping frontend build.

============================================================
 6. Dependencies: MSz and FFCz
============================================================
[INFO] MSz is ALREADY INSTALLED and importable in Python.
Skip building MSz? [Y/n]
[INFO] Skipping MSz build.
[INFO] FFCz is ALREADY INSTALLED (/home/guoxil/software/spack/
Skip building FFCz? [Y/n]
[INFO] Skipping FFCz build.

============================================================
 7. Generating Runner Script
============================================================
[INFO] Configuration memorized in /home/guoxil/software/spack/
```

Install FZ-VIS via Spack (Method 2)

- Download spack packages for FZ

```
$ git clone https://github.com/guoxiliu/spack_packages.git
```

- Add spack packages

```
$ spack repo add spack_repo/robertu94
$ spack repo list
```

- Install FZ-VIS using spack

```
$ spack install fzvis
```

Run FZ-VIS Locally

- Run the installation script (from method 1)

  ```
  $ ./scripts/run_fzvis.sh
  ```

- Open the following URL in a web browser

  ```
  http://localhost:10080
  ```

```
============================================================
FZ-VIS installation complete!
============================================================
Active environment: /home/guoxil/software/spack/var/spack/environments/fzvis-env

Component Verification Status:
  - MSz (Python module):       INSTALLED & VERIFIED
  - FFCz (binary executable):  INSTALLED & VERIFIED
  - Frontend (dist/):          BUILT & READY

Start the application with:
  /home/guoxil/software/spack/var/spack/environments/fzvis-env/fzvis/scripts/run_fzvis.sh

Then access in browser at:
  http://localhost:10080
```

Configure FZ-VIS Parameters

- Copy the `.env.example` and rename it `.env`
- Adjust the following parameters as needed
  - passcode to access FZ-VIS

```
FLASK_PASSCODE=‘’
```

  - the number of input/decompressed data to be kept in the cache

```
FZVIS_INPUT_CACHE_SIZE=50
FZVIS_DECOMPRESSED_CACHE_SIZE=50
```

  - maximum number of compressors to run by each user

```
FZVIS_MAX_COMPRESSOR_INSTANCES=10
```

  - file upload size limit

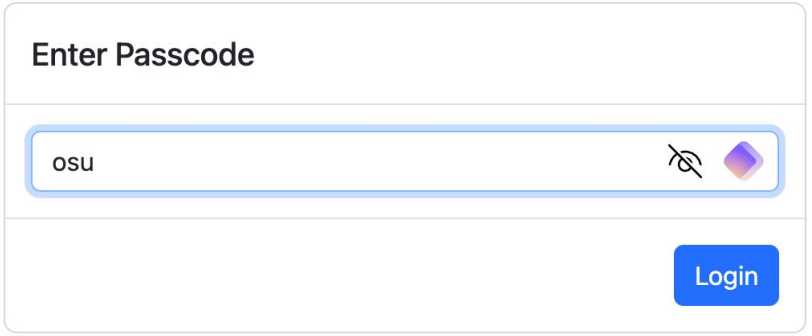
```
FZVIS_MAX_UPLOAD_SIZE='100MB’
```

Deploy FZ-VIS

- The built-in `run_fzvis.sh` serves the static production through the port 10080.
  - `http://<public_ip>:10080`
- Use a web server for reverse proxy (e.g., Apache, NGINX)
  - If you want to use standard 80/443 port
  - Run `$npm run build` to get the production build in `dist` folder
  - Copy files from `dist` folder to `/var/www/html` folder

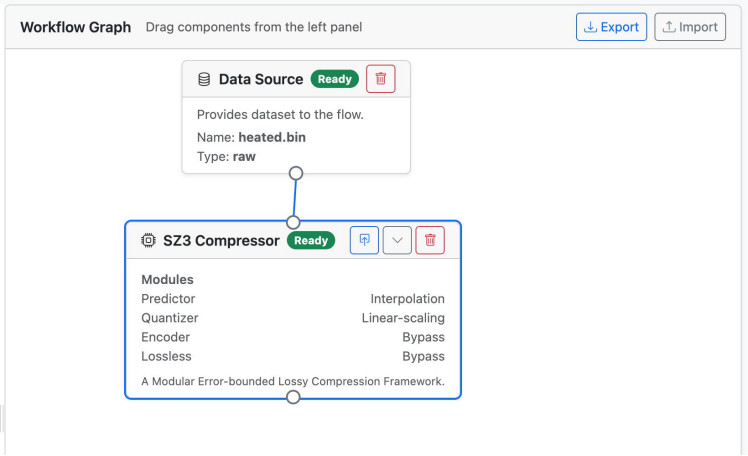
FZ-VIS Live Demo

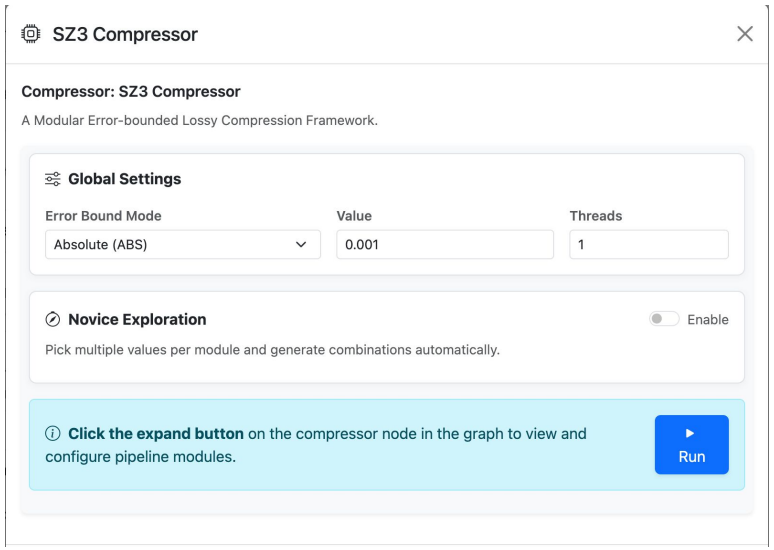
- URL: https://go.osu.edu/fzvis or http://20.106.237.44/
- Passcode: osu
- Chameleon instance: https://go.osu.edu/fzvis1

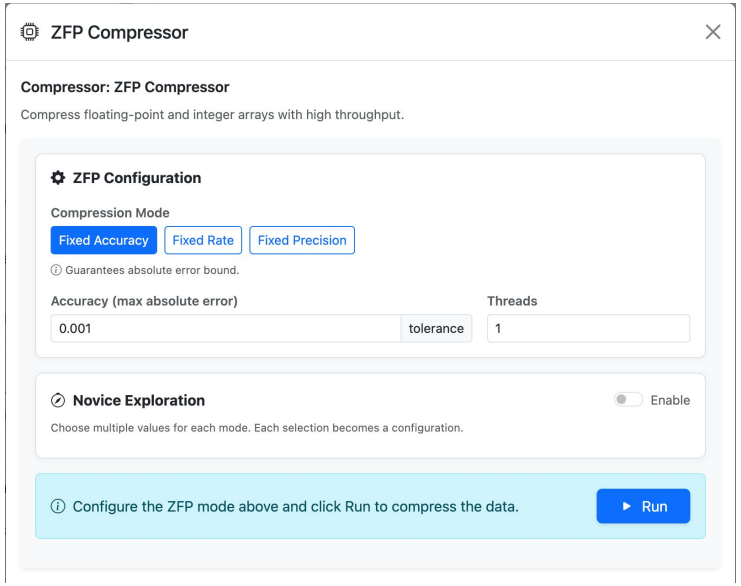


Using Existing Compressors in FZ-VIS

- Double click to open global settings of the compressor

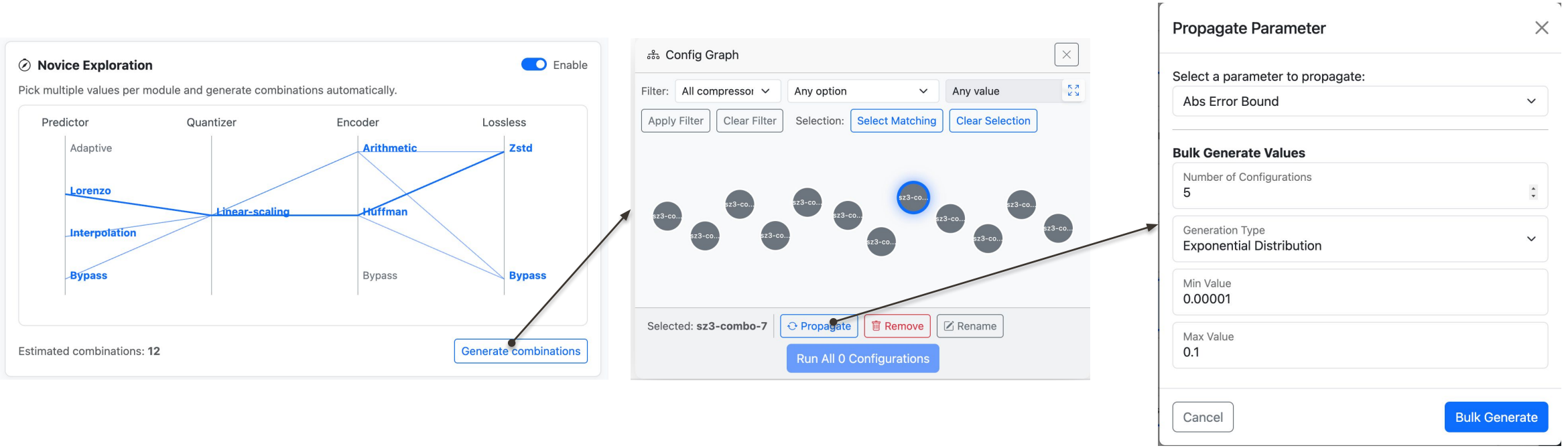






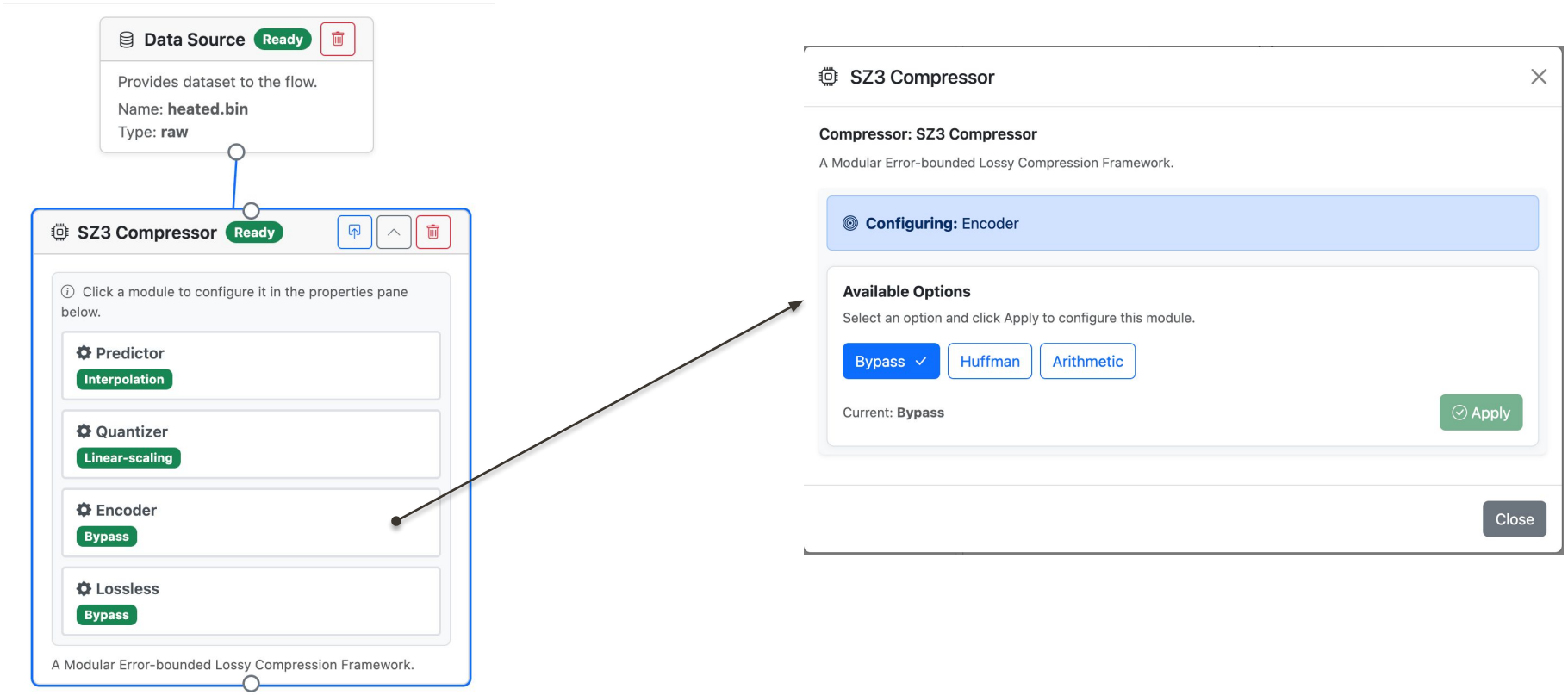
Using Existing Compressors in FZ-VIS

- Use parallel coordinates and config graph to quickly generate instances



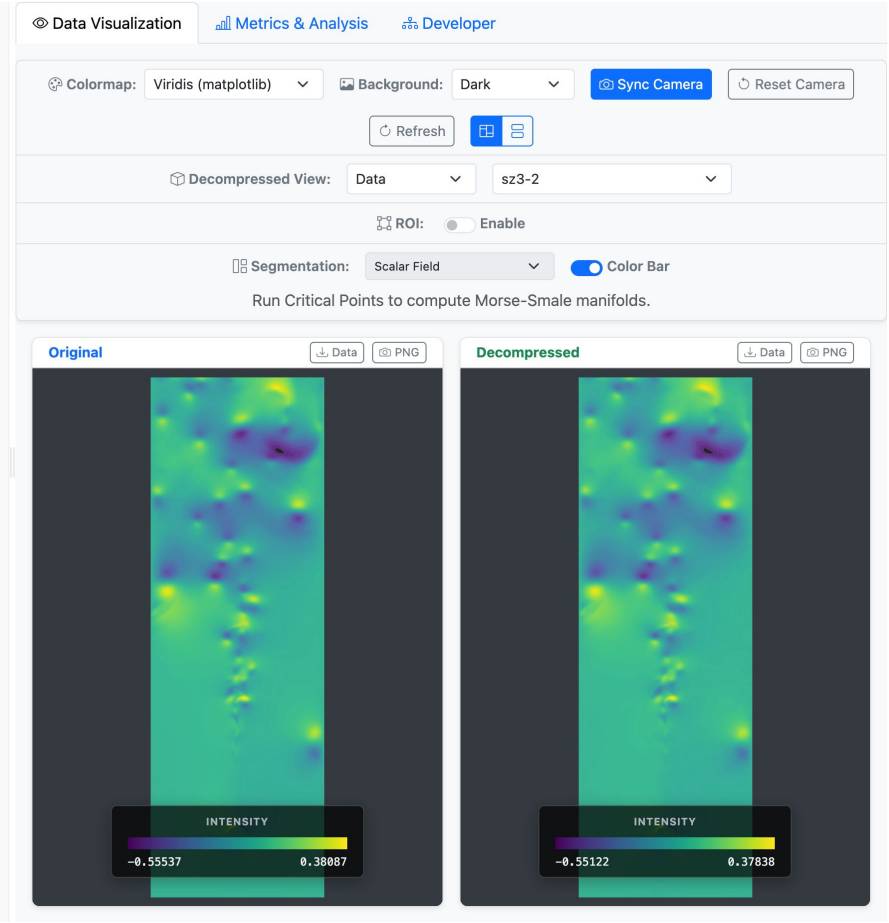
Creating Compressors with Modules in FZ-VIS

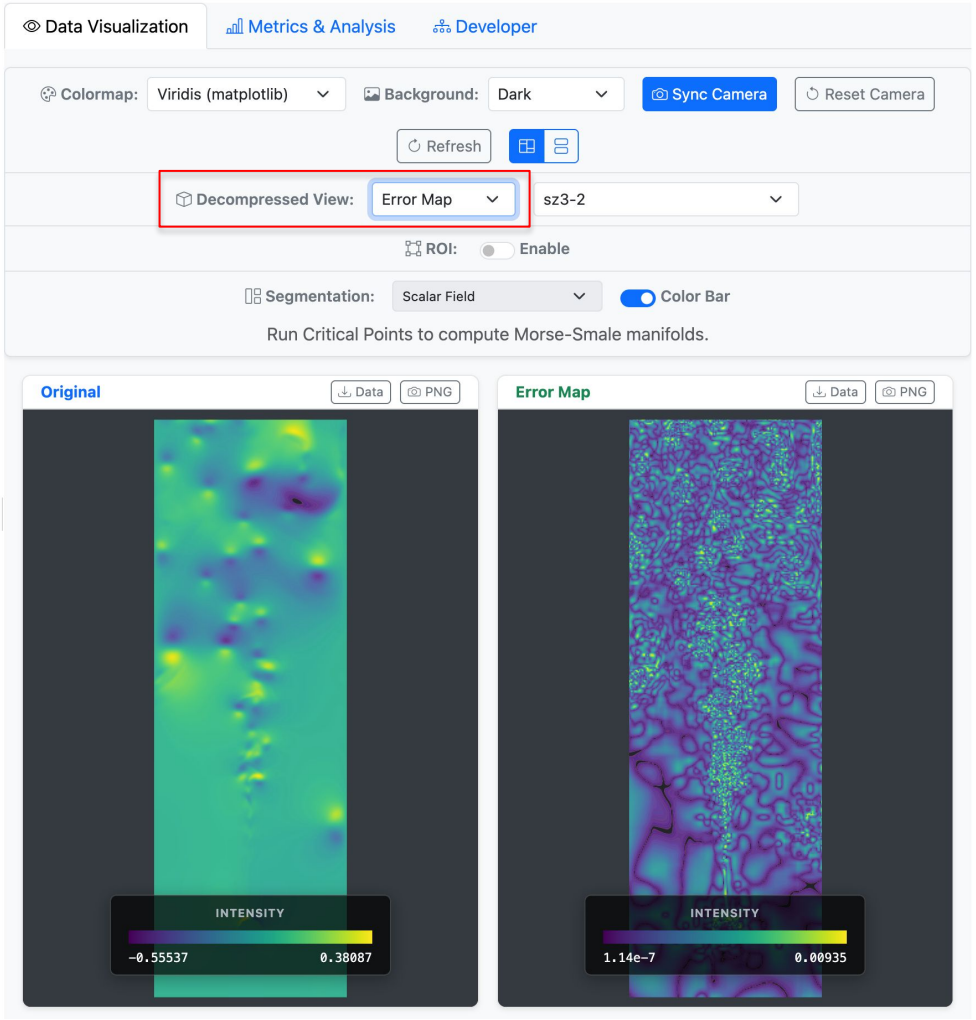
- Expand to explore the modules of the compressor



Visual Quality Analysis in FZ-VIS

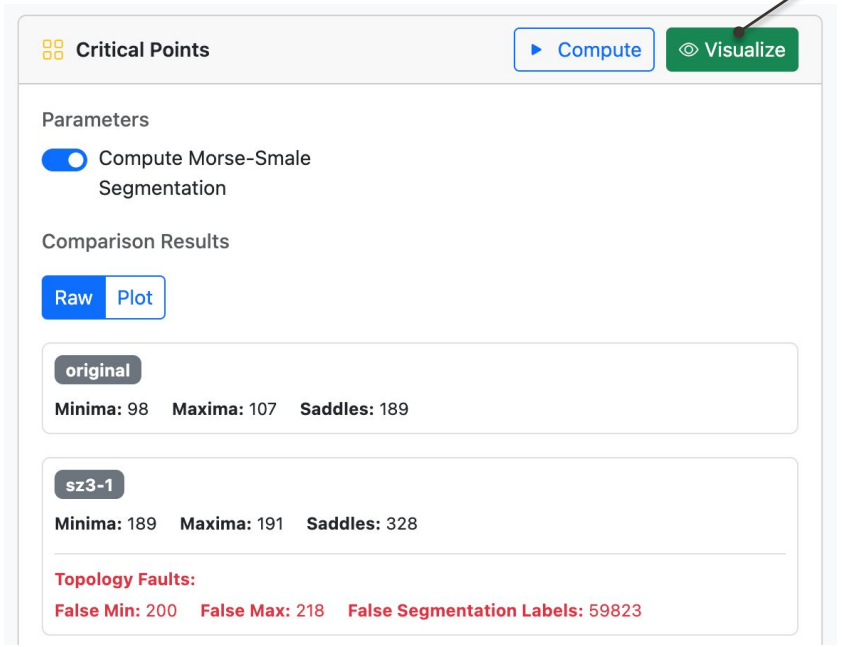
- Spatial comparison and error map

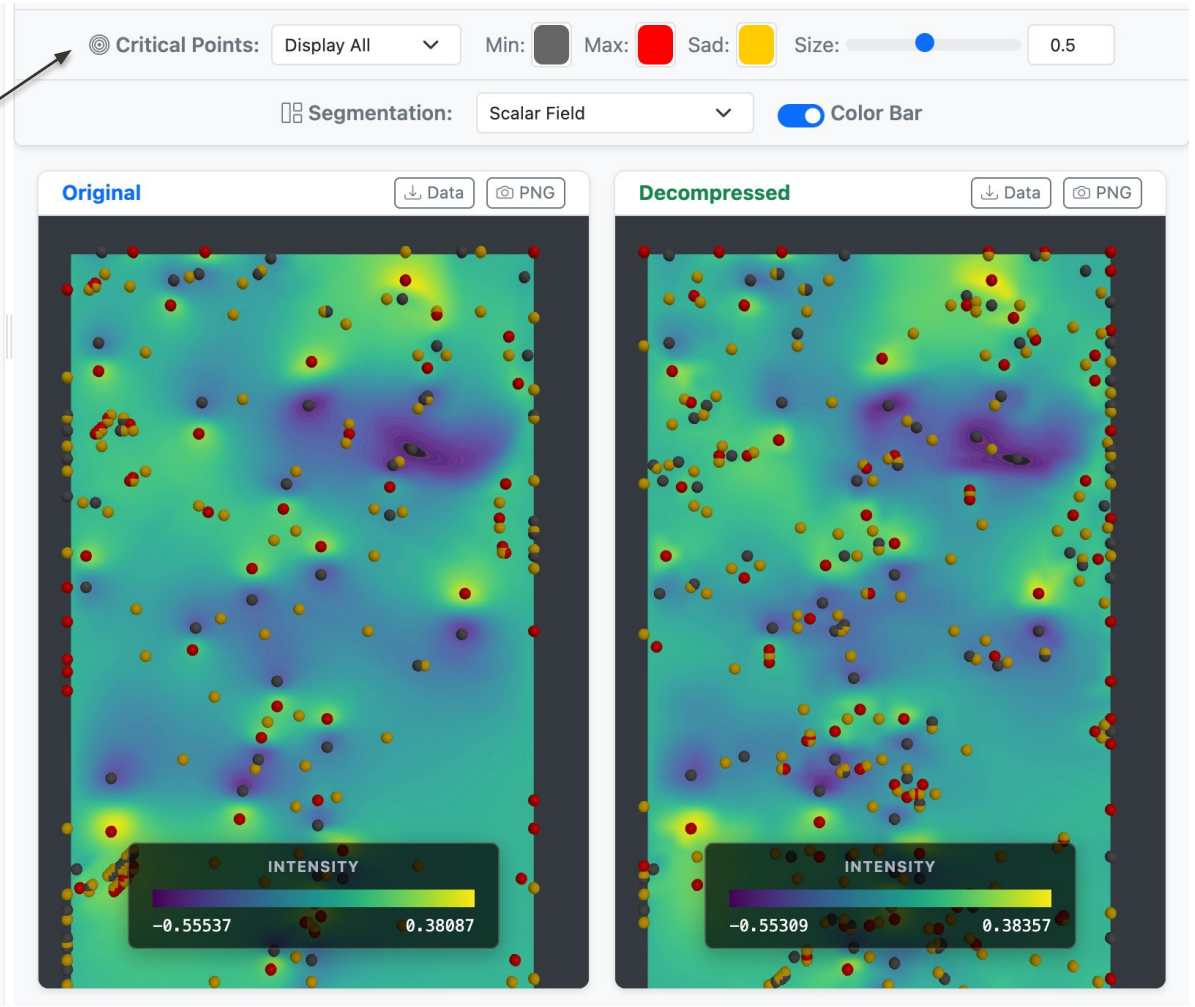




Quantities of Interests Analysis in FZ-VIS

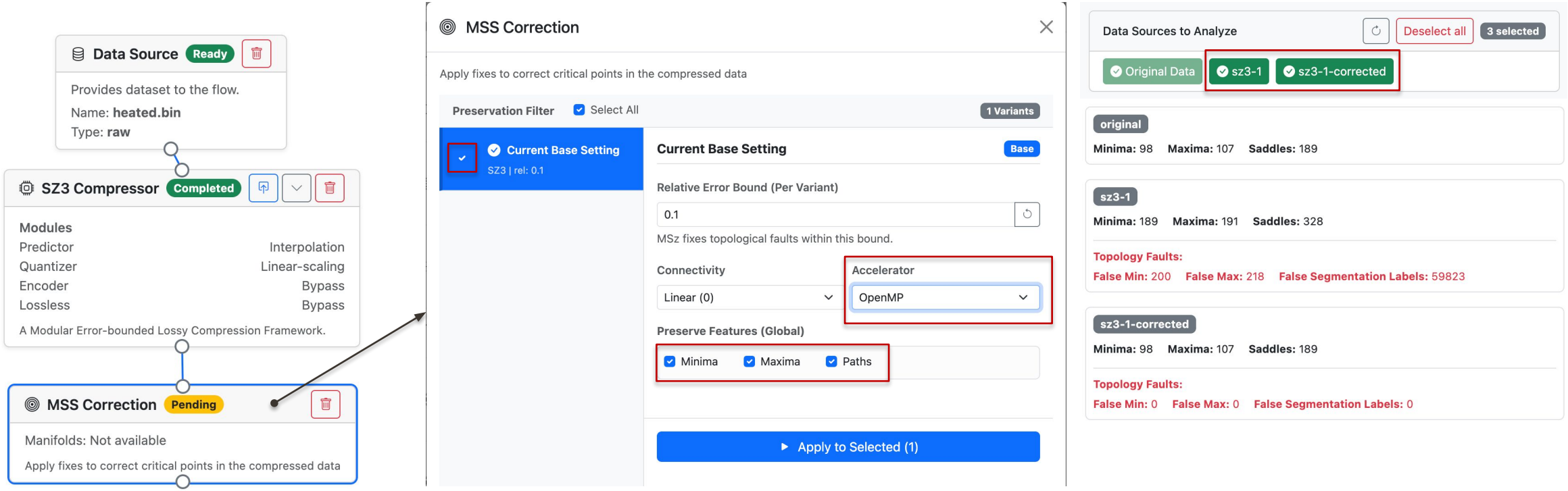
- Topological domain





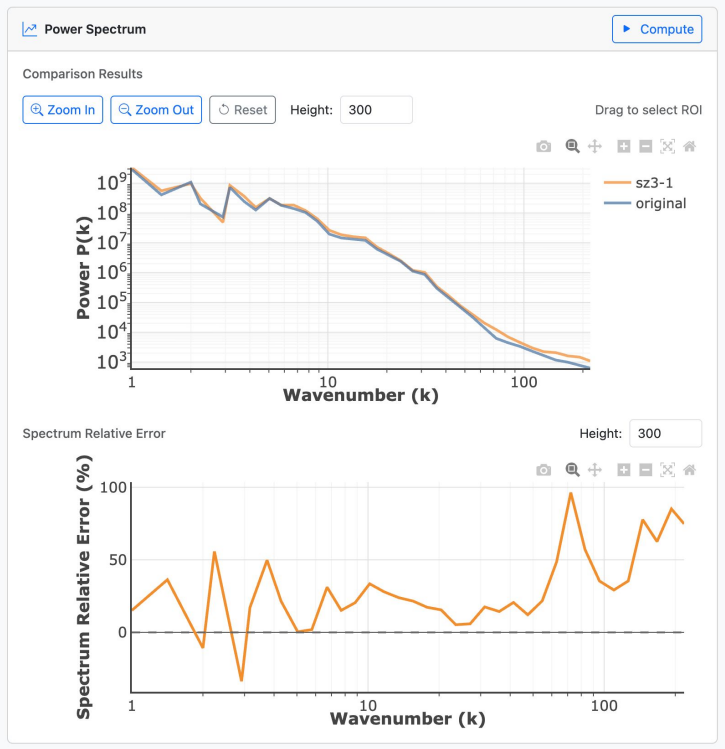
Quantities of Interests Analysis in FZ-VIS

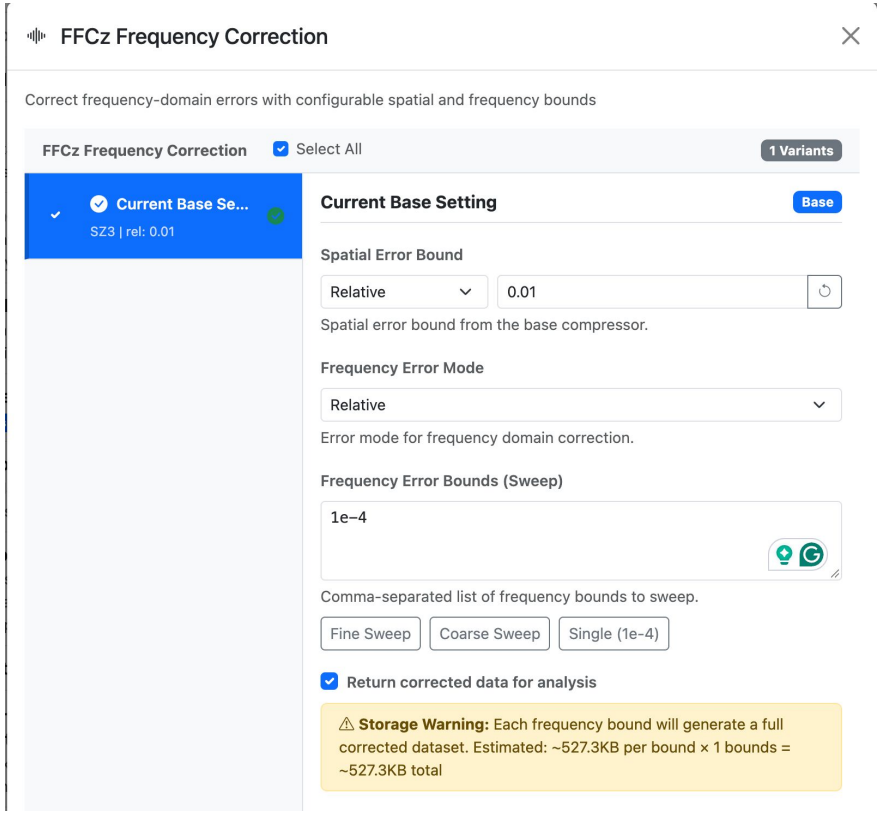
- Topological domain

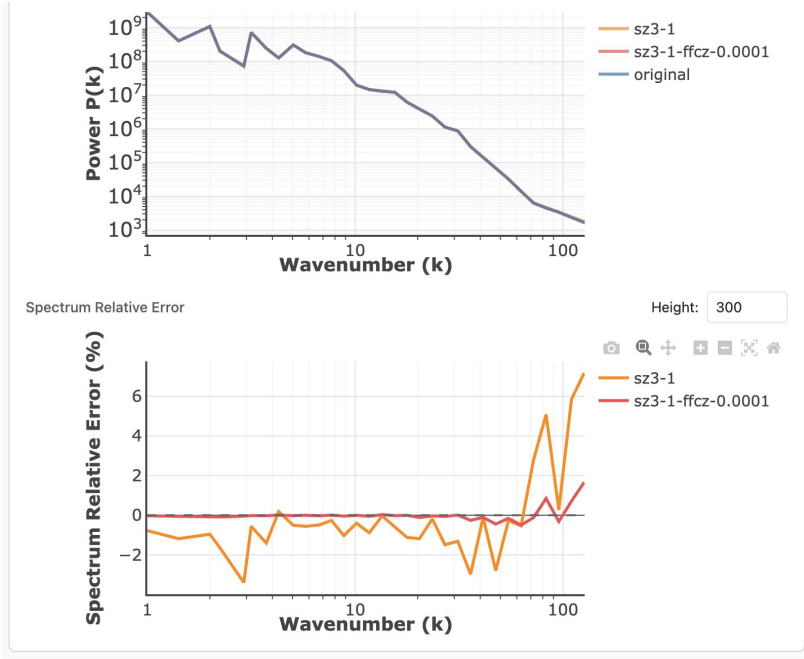


Quantities of Interests Analysis in FZ-VIS

- Frequency domain







Thank you!

Questions? liu.12722@osu.edu

GitHub repo: https://github.com/YuxiaoLi1234/fzvis





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FZ-VIS: A Visual Analytics Framework for Quantities-of-Interest-Aware Scientific Lossy Compression

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